4E1301

Roll No. ____________________ [Total No. of Pages : 3

4E1301

B.Tech. IV Sem. (Main/Back) Examination, July - 2023
Artificial Intelligence and Data Science
4AID2-01 Discrete Mathematics Structure
CS, IT, AID, CAI

Time : 3 Hours **Maximum Marks : 70**

Instructions to Candidates:

*Attempt **all ten** questions from **Part A**, Attempt any five questions out of **Seven** questions from **Part B** and **three** questions out of **Five** questions from **Part C**.*

Schematic diagrams must be shown wherever necessary. Any data missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (As Mentioned in form No. 205).

PART - A

(Word limit 25)

All questions are compulsory. **(10×2=20)**

1. Prove that $A - B = B' \cap A'$. **(2)**
2. Define partial ordering relation with an example. **(2)**
3. Draw the truth table for $p \wedge q \rightarrow p \vee q$. **(2)**
4. Explain conjuctive normal form. **(2)**
5. Draw the Hasse diagram of the poset $(A, \leq)$ where A = {1,2,3,4,12} and the partial order of divisibility on A is $a \leq b$ (i.e. if a divides b). **(2)**
6. 8 boys and 5 girls constitute a group. In how many ways seven of them can be selected if the selections always have atleast 3 boys and 2 girls. **(2)**
7. If a,b are any elements of a group G, then prove that $(ab)^{-1} = b^{-1}a^{-1}$. **(2)**
8. If a is a generator of a cyclic group, then prove that a^{-1} is also its generator. **(2)**
9. Draw graph which is
   a. Eulerian but not Hamiltonian.
   b. Hamiltonian but not Eulerian. **(2)**
10. Define chromatic number. **(2)**

PART - B

(Word limit 100)

Attempt any Five questions. **(5×4=20)**

1. In a class of 60 students, 25 study Hindi, 26 study English and 26 study Sanskrit. Also 9 study both Hindi and Sanskrit, 11 study hindi and english and 8 study Englishh and Sanskrit. If 8 study none of the three subjects, Find number of students who study exactly one subject. **(4)**

2. Using principle of mathematical induction, prove that $1+2+2^2+.........+2^{n-1}=2^n-1$. **(4)**

3. Draw the transition diagram of finite state machine represented by following state table. Also find output word corresponding to input word w = 11011011. **(4)**

State	Transition	function	Output	function
	0	1	0	1
s_0	s_3	s_1	0	1
s_1	s_0	s_1	0	1
s_2	s_3	s_1	0	1
s_3	s_1	s_3	0	0

4. Solve the recurrence relation

   $a_r - 4a_{r-1} + 4a_{r-2} = 0$; $a_0 = 1$, $a_1 = 3$. **(4)**

5. Define isomorphism of groups. Prove that every subgroup of an abelian group is normal. **(4)**

6. Prove that the sum of degrees of all the vertices in a graph is equal to twice the number of edges in the graph. **(4)**

7. Prove that set of real numbers of the form $a+b\sqrt{2}$ (where a and b are integers) with ordinary addition and multiplication forms a ring. **(4)**

PART - C

Attempt any Three questions. **(3×10=30)**

1. In the set Z of integers, a relation R is defined by $aRb \Leftrightarrow a \equiv b(\text{mod } 4)$. Show that R is an equivalence relation. **(10)**

2. Define universal and existential quantifiers. prove that $p \to (q \wedge r) \equiv (p \to q) \wedge (p \to r)$. **(10)**

3. Prove that the dual of a lattice is also a lattice. **(10)**

4. Prove that the necessary and sufficient conditions for a non - void subset H of a group G to be a subgroup is that $a \in H, b \in H \Rightarrow ab^{-1} \in H$. **(10)**

5. Find the shortest path between the vertices v_0 and v_5 in the following weighed graph. **(10)**